

Using causal matching to understand the effects of misinformation on social media users

While misinformation accounts for only a small fraction of shared content (Allen et al., 2020; González-Bailón et al., 2024), its effects on public discourse remain debated (Altay, 2026; Budak et al., 2024). Existing studies often face two limitations: they rely on biased samples of clearly false news, ignoring relevant gray-area content (Allen et al., 2024); and most evidence comes from correlational observational data or experiments using artificial stimuli and science-friendly samples, unable to capture real users encountering real misinformation. Deliberately exposing participants to misinformation also raises ethical concerns about informed consent and lasting harm (Lazer et al., 2021). Several researchers therefore call for rethinking causality in this context (Lorenz-Spreen et al., 2022; Tay et al., 2024), developing designs better suited to the complex dynamics of social media (Bak-Coleman et al., 2025), and complementing the existing evidence with meaningful estimates of its effects *in the wild*.

Large-scale social media data has huge potential for studying how misinformation spreads in a naturalistic environment but introduces confounding challenges. Purely observational designs struggle to disentangle misinformation effects from co-occurring confounders like political polarization, hate speech and user self-selection, making it difficult to capture the emergent dynamics that arise from users, algorithms, and content (Bak-Coleman et al., 2025). Causal inference via non-parametric matching preserves the naturalistic context while blocking these structural confounders: rather than randomly assigning users, matching creates a synthetic control group that is statistically comparable to the treatment group on observed confounders, thereby blocking spurious associations between treatment and outcome (so-called backdoor paths), enabling causal estimation where experimental manipulation is infeasible or unethical (Ho et al., 2007; Pearl & Mackenzie, 2019). We applied this approach to a large-scale dataset of 9.9M news tweets and 11M replies on Twitter (October 2020–March 2022) spanning German news sources of varying trustworthiness, matching on political bias, tweet-level features (word count, emotion), and author characteristics (follower, following, and tweet counts). To measure emotional responses, we applied a transformer-based classifier to identify seven discrete emotions in replies, separating emotions in the news stimulus from those in the replies to control for emotional carryover.

Despite comprising only 6% of shared news, untrustworthy sources receive 39% more retweets and 12% more quote retweets, but 14% fewer likes and 19% fewer replies compared to matched trustworthy sources (Figure 1a). This divergent pattern suggests selective amplification rather than viral engagement, consistent with findings that removing reshare buttons reduces untrustworthy news in feeds (Guess et al., 2023). In the discussions, replies to untrustworthy sources express 9% more anger, 22% more disgust, 12% more fear, and 12% less joy relative to matched trustworthy sources (Figure 1b). This pattern was also present in only the first replies, ruling out emotional contagion as a driver. These findings extend prior work showing that misinformation exploits moral outrage (McLoughlin et al., 2024) and high-arousal negative emotions (Brady et al., 2017). Higher anger is partly driven by self-selection: within-person comparisons restricted to users who replied to both source types, show that the same user does not express more anger in response to untrustworthy content; rather, angrier users preferentially engage with it. Consistent with that, only 11% of users ever replied to untrustworthy sources, consistent with selective engagement by a responsive minority. By contrast, disgust and fear show genuine within-person increases.

These findings reveal distinct fingerprints of misinformation: active resharing rather than liking, combined with elevated anger, fear, and disgust, driven by selective participation from emotionally reactive users. Methodologically, the study disentangles treatment effects from user self-selection in data where randomization is infeasible. Our causal inference design combined with within-user comparisons shows that selection is not only a confounder to be controlled, but a substantive part of how misinformation spreads through a small, responsive minority. More broadly, this approach illustrates how causal inference paired with individual-level sensitivity analyses can address the validity and generalizability concerns of platform data, offering a template for media effects research where experimental randomization is infeasible or unethical.

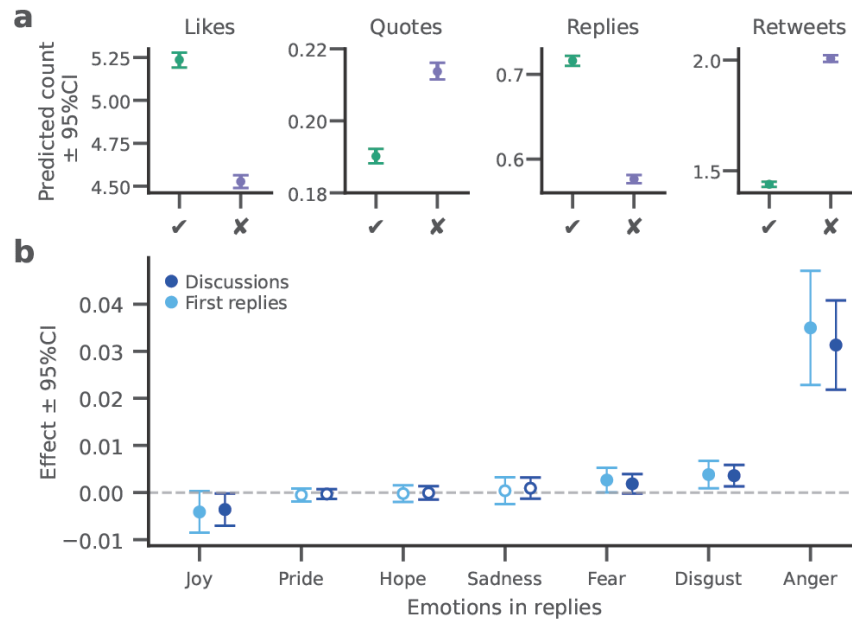


Figure 1

Panel a: Predicted counts of engagement (retweets, quote retweets, likes, replies) per condition with 95% confidence intervals. Panel b: Beta coefficients for effects on discrete emotion intensities in discussions and first replies with bootstrapped 95% confidence intervals. Green = trustworthy; purple = untrustworthy.

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